Definition 23 (Update).
For $D \in C$, but with all arbitrary objects novelised,
$\langle C, \Gamma^{\Theta}_{RI}\rangle[D]^{\circlearrowright} = \langle C, \Gamma^{\Theta}_{RI}[D]^{U}[D]^{E}[D]^{A}[D]^{M}\rangle$

Definition 24 (Universal product).
For $\mathbf{A}(\Gamma) \cap \mathbf{A}(\Theta) = \emptyset$ and either $\mathbf{A}(R) \cap \mathbf{A}(S) = \emptyset$ or $[R]_{\Delta} = S$,
$\Gamma^{\Theta}_{RI}[\Delta^{\{0\}}_{SJ}]^{U} = \{0\}^{\Theta}_{RI} \otimes^{R\rhd\lhd S} \bigotimes^{R\rhd\lhd S}_{\langle u,t\rangle \in M'_{IJ}} \mathrm{Sub}^{R\rhd\lhd S}_{\langle t,u\rangle}(\Gamma^{\{0\}}_{I})$, *where*
$M'_{IJ} = \{\langle u,t\rangle : \langle u,t\rangle \in M_{IJ} \wedge u \in U_R - \mathbf{A}(\Theta)\} \neq \emptyset$

Definition 25 (Reorient).
For $\Delta^{\Psi}_{SJ} \in (C \cup \{\Gamma^{\Theta}_{RI}\})$ and $J' \in \mathbb{I}_{\Delta\cup\Psi}$,
$\langle C, \Gamma^{\Theta}_{RI}\rangle[\Delta^{\Psi}_{SJ'}]^{R} = \langle C, \Delta^{\Psi}_{SJ'}\rangle$

Definition 26 (Existential sum).
Let the following conditions be met:

(1) $\mathbf{A}(\Gamma) \cap \mathbf{A}(\Theta) = \emptyset$ *and either* $\mathbf{A}(R) \cap \mathbf{A}(S) = \emptyset$ *or* $[R]_{\Delta} = S$.

(2) $M'_{IJ} = \{\langle e,t\rangle : \langle e,t\rangle \in M_{IJ} \wedge e \in E_R - \mathbf{A}(\Theta \cup \Delta) \wedge \neg\exists x(\langle e,x\rangle \in D_R)\} \neq \emptyset$.

Then,

$$\Gamma^{\Theta}_{RI}[\Delta^{\{0\}}_{SJ}]^{E} = \Gamma^{\Theta}_{RI} \oplus^{R\rhd\lhd S} \bigoplus^{R\rhd\lhd S}_{\langle e,t\rangle \in M'_{IJ}} \mathrm{Sub}^{R\rhd\lhd S}_{\langle t,e\rangle}\Big(\Big(\bigcup_{\gamma \in \{\gamma_{\in\Gamma} : e \in \mathbf{A}(\gamma)\}} (\{\gamma\} \cup \big\{\{x_{\in\gamma} : e \notin \mathbf{A}(x)\} \cup \delta : \delta \in \bigotimes_{x \in B(\gamma,I,e)} \{\{x\},\{\bar{x}\}\} \wedge \delta \not\subseteq \gamma\big\})\Big)^{\Theta}_{I}\Big),$$

where for $e \in E_R$ and $\gamma \in \Gamma$, let $B(\gamma, I, e) = \{x[e/?] \in \gamma : \langle e,x\rangle \in I\}$.

Definition 27 (Division).
If $\forall\delta_{\in\Delta}\exists\psi_{\in\Psi}\exists\gamma_{\in\Gamma}(\delta \subseteq \gamma \wedge \psi \subseteq \gamma)$, then
$\Gamma^{\Theta}_{RI} \oslash \Delta^{\Psi}_{SJ} = \{\gamma \oslash_{\Gamma} \Delta^{\Psi} : \gamma \in \Gamma\}^{\Theta}_{[R][I]}$
$\gamma \oslash_{\Gamma} \Delta^{\Psi} = \gamma - \iota\delta(\delta \in \Delta \wedge \delta \subseteq \gamma \wedge \exists\psi_{\in\Psi}(\psi \subseteq \gamma))$

Definition 28 (Factor).
For $\Delta^{\Psi}_{SJ} \neq \bot$,
$\Gamma^{\Theta}_{RI}[\Delta^{\Psi}_{SJ}]^{F} = \{\gamma[\Delta^{\Psi}]^{F} : \gamma \in \Gamma\}^{\Theta}_{[R][I]}$ *where*
$\gamma[\Delta^{\Psi}]^{F} = (\gamma \oslash_{\Gamma} \Delta^{\Psi}) \cap \bigcap\{\gamma \oslash_{\Gamma} (\Delta^{\Psi}[t/a]) : \langle t,a\rangle \in M_{IJ} \wedge a \in U_S\}$, *and where we let* $\gamma' \cap \bigcap\emptyset = \gamma'$.
For $\Delta^{\Psi}_{SJ} = \bot$,
$\Gamma^{\Theta}_{RI}[\bot]^{F} = \big\{\gamma \in \Gamma : \neg\exists\delta \in \mathbb{K}.\delta \subseteq \gamma\big\}^{\Theta}_{[R][I]}$.

Definition 29 (Matryoshka level ordering)**.**
For existentials, we have $e \lesssim_R e'$ iff $\forall\langle e',u\rangle \in D_R(\langle e,u\rangle \in D_R)$. For both existentials and universals, let the relation $\lhd_R$ be the transitive closure of the relation made up by the ordered pairs in $D_R \cup (E_R \times U_R - D_R)^{\mathrm{op}}$, where op reverses the order of pairs in a set of pairs.

Definition 30 (Query).
For $U_S \subseteq U_R$ and $M'_{IJ} = \{\langle t,e\rangle \in M_{IJ} : e \in E_S \backslash E_R\}$, we define
$\Gamma^{\Theta}_{RI}[\Delta^{\Psi}_{SJ}]^{Q} =$

$$(\{0 : \neg\exists\delta_{\in\Delta}\exists\gamma.\Phi(\gamma,\delta)\} \cup \{\delta \in \Delta : \exists\gamma_{\in\Gamma}.\Phi(\gamma,\delta)\})^{\Theta}_{[R\rhd\lhd\langle U_R, E_S\backslash E_R, D_{S'}\rangle][I\cup J]}$$

where $\Phi(\gamma,\delta) \leftrightarrow \exists\psi_{\in\Psi}\exists n_{\geq 0}\exists\langle t_1,e_1\rangle,\ldots,\langle t_n,e_n\rangle_{\in M'_{IJ}}\forall i,j$
$(\psi \cup \delta[t_1/e_1,\ldots,t_n/e_n] \subseteq \gamma \wedge e_i = e_j \rightarrow i = j)$;
and $D_{S'} = D_1 \cup D_2 \cup D_3 \cup D_4 \cup D_5 \cup D_6$ is constructed by taking: original dependencies from the internal argument

$$D_1 = [D_S]_{\mathbf{A}/E_R},$$

extra dependencies for multiple terms substituted with same e

$$D_2 = \{\langle e_m,u\rangle : \exists m\exists m' \in M'_{IJ}(e_m = e_{m'} \wedge t_m \neq t_{m'}) \wedge u \in U_R\},$$

dependencies resulting from complex terms being substituted

$$D_3 = \{\langle e_m, u\rangle : \langle t_m, e_m\rangle \in M'_{IJ} \wedge u \in U_R(t_m)\},$$
$$D_4 = \{\langle e_m, u\rangle : \langle t_m, e_m\rangle \in M'_{IJ} \wedge e \in E_R(t_m) \wedge \langle e, u\rangle \in R\},$$
$$D_5 = \{\langle e_m, u\rangle : \langle t_m, e_m\rangle \in M'_{IJ} \wedge u \in U_R \wedge \forall u'_{\in U_R(D_3 \cup D_4)}(u' \lhd_R u)\},$$

additional dependencies necessary to preserve the dependency order of existentials in S

$$D_6 = \{\langle e, u\rangle : e, e' \in E_S - E_R \wedge e \lesssim_S e' \wedge$$
$$(\forall m, m' \in M'_{IJ}(e_m = e' \wedge e_{m'} = e') \rightarrow t_m = t_{m'}) \wedge \langle e', u\rangle \in D_1 \cup D_2 \cup D_3 \cup D_4 \cup D_5\}.$$

Definition 31 (**Wh-Query**). *Provided that $U_S \subseteq U_R$,*
$\Gamma^{\Theta}_{RI}[\Delta^{\Psi}_{SJ}]^W = \Big(\big\{0 : \exists\gamma \in \Gamma.\neg\exists\delta_{\in\Delta}.\Phi(\gamma,\delta)\big\} \cup \big\{\delta \in \Delta : \exists\gamma_{\in\Gamma}.\Psi(\gamma,\xi)\big\}\Big)^{\Theta}_{[R][I]}$, *where $\Psi(\gamma,\xi)$ iff*

$$\exists\psi_{\in\Psi}\exists n_{\geq 0}\exists\langle t_1, e_1\rangle, \ldots, \langle t_n, e_n\rangle_{\in M'_{IJ}}\exists\delta_{\in\Delta}.$$
$$\big(\xi \cup \psi \subseteq \gamma \wedge \xi = \delta[t_1/e_1, \ldots, t_n/e_n]$$
$$\wedge\, \forall i, j.(e_i = e_j \rightarrow i = j)\big).$$

Definition 32 (**Inquire**).
(O) If $\mathbf{A}(\Gamma \cup \Theta) \cap \mathbf{A}(\Delta \cup \Psi) = \emptyset \wedge \mathbf{A}(\Delta) \cap \mathbf{A}(\Psi) = \emptyset$, then
$\Gamma^{\Theta}_{RI}[\Delta^{\Psi}_{SJ}]^I = \Gamma^{\Theta}_{RI} \otimes \Big(\Delta^{\Psi}_{SJ} \oplus \big(\{0\}^{\Psi}_{[S][J]} \otimes Nov([\Delta^{\{0\}}]^N_{[[S]^N][J]^N})\big)\Big)[\bot]^F$,
where $Nov()$ uniformly replaces all arbitrary objects in its scope by novel ones.

(I) If $\mathbf{A}(\Delta \cup \Psi) \subseteq \mathbf{A}(\Gamma \cup \Theta)$, then
$\Gamma^{\Theta}_{RI}[\Delta^{\Psi}_{SJ}]^I = \Gamma^{\Theta}_{RI} \otimes \big(\Delta^{\Psi}_{0_{\mathcal{R}}J} \oplus ([\Delta]^N_{0_{\mathcal{R}}[J]^N})^{\Psi}\big)[\bot]^F$.

Definition 33 (**Suppose**).
(O) If $\mathbf{A}(\Gamma \cup \Theta) \cap \mathbf{A}(\Delta \cup \Psi) = \emptyset$, then
$\Gamma^{\Theta}_{RI}[\Delta^{\Psi}_{SJ}]^S = \Gamma^{\Theta'}_{[R\bowtie R'][I\cup I']}[\Delta^{\Psi}_{SJ}]^U[\Delta^{\Psi}_{SJ}]^E[\Delta^{\Psi}_{SJ}]^A[\Delta^{\Psi}_{SJ}]^M$, *where*
$\Theta'^{\{0\}}_{R'I'} = \Theta^{\{0\}}_{RI} \otimes Nov(\Delta^{\Psi}_{[S]^N J}[\,]^D)$,
where $Nov()$ uniformly replaces all arbitrary objects in its scope by novel ones.

(I) If $\mathbf{A}(\Delta) \subseteq \mathbf{A}(\Gamma \cup \Theta)$ and $[R]_{\Delta} = S$,
then $\Gamma^{\Theta}_{RI}[\Delta^{\{0\}}_{SJ}]^S = \Gamma^{\Theta\otimes\Delta}_{RI}[\Delta^{\{0\}}_{SJ}]^U[\Delta^{\{0\}}_{SJ}]^E[\Delta^{\{0\}}_{SJ}]^A[\Delta^{\{0\}}_{SJ}]^M$

Definition 34 (**Depose**).
$\Gamma^{\Theta}_{RI}[\top]^D = \langle\Gamma^{\{0\}} \oplus ([\Theta]^N)^{\{0\}}, R, [[I]^N \cup I]\rangle$.

Definition 35 (**Commit**).
$\langle C, \Gamma^{\Theta}_{RI}\rangle[\top]^C = \langle C \cup \{\Gamma^{\Theta}_{RI}\}, \Gamma^{\Theta}_{RI}\rangle$

Definition 36 (Inference)**.** *An inference $\mathcal{I}$ is a finite sequence of inference steps defined as follows, relative to an inference state consisting of a commitment $\langle C, G\rangle \in \mathbb{C}$. If $E \in \mathbb{V}$ and $O \in \{[D]^{\circlearrowleft}, [D]^S, [D]^Q, [D]^F, [D]^R, [D]^I, [D]^D, [D]^E, [D]^U, [D]^C\}$, then O is an inference step applicable to $\langle C, G\rangle$, whose result is an inference state in $\mathbb{C}$. We write $C \vdash_{\mathrm{ETR}} E$ if and only if there is an erotetic theory inference $\mathcal{I}$ s.t. $\langle C, \top\rangle\mathcal{I} = \langle C', E'\rangle$, where E' is identical to E up to uniform replacement of $\mathbf{A}$ objects.*